# POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2024
Programme: BE | Full Marks : 100
Course: Numerical Methods (New) | Pass Marks : 45
Time : 3 hrs.

*Candidates are required to give their answers in their own words as far as practicable.*

*The figures in the margin indicate full marks.*

***Attempt all the questions.***

1. a) Solve $x \log_{10} x = 1.2$ by Newton- Raphson method correct to four decimal places. 8

b) Using Secant Method, find the roots of function $2x - \log_{10} x - 7 = 0$ Correct up to three decimal places. 7

**OR**

Find the root of the equation $f(x) = x^2-4x-10$ correct to three decimal places by using False Position method.

2. a) From the following data given in the table below evaluate f (2.5) by using Lagrange method. 8

| x | 1 | 2 | 4 | 5 | 7 |
|---|---|---|---|---|---|
| f(x) | 1 | 1.414 | 1.732 | 2.00 | 2.6 |

b) From the following table, Estimate the number of student who obtained marks between 50 and 55. 7

| Marks | 30-40 | 40-50 | 50-60 | 60-70 | 70-80 |
|---|---|---|---|---|---|
| No of Students | 31 | 42 | 51 | 35 | 31 |

3. a) Compute the Simpson's 1/3 and Simpson's 3/8 rule for $I = \int_0^1 e^{-x^2} dx$ using a regular partition with subinterval n=6. 8

b) Use the Romberg integration to find the solution correct upto three decimal places. 7

$$I = \int_0^1 \frac{1}{1+x^2}$$



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4. $3x+2y+z=10$ 8

a) Solve : $2x+3y+2z=14$

$x+2y+3z=14$

By Gauss elimination method.

b) Solve the following system of equations using Crout method. 7

$x+y+z=4, x+4y+3z=8, x+6y+2z=6$

**OR**

Find largest eigen value and corresponding eigen vector of the matrix.

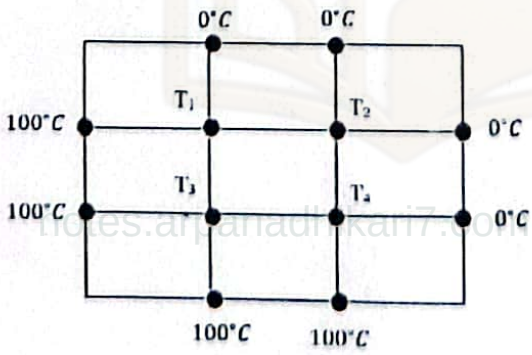
$$\begin{bmatrix} 3 & -1 & 0 \\ -2 & 4 & -3 \\ 0 & -1 & 1 \end{bmatrix}$$

5. a) Solve the following differential equation within $0 \le x \le 0.3$ using RK-4$^{TH}$ order Method. 8

$$10\frac{dy}{dx} = x^2 + y^2, y(0) = 1 \; with \; h = 0.1$$

b) Apply Euler's method to approximate value of $y(0.3)$ for the differential equation: $\frac{dy}{dx} = y + x, y(0) = 1$. 7

6. a) For square bar of size 15cm×15cm, calculate the steady state temperature at interior point for the grid size of 5cm×5cm. 8




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b) Solve the Poisson equation $\nabla^2 f = 2x^2 + y$, over the square domain $1 \le x \le 4$, $1 \le y \le 4$, with $f=0$ on the boundary. Take step size in x and y, $h=k=1$. 7

7. Write short notes on: **(Any two)** 2×5

a) Errors in Numerical Method

b) Ill-conditioned systems

c) Boundary value problem



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